

Event Collision Detector

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1. Introduction

In today's hyper-connected landscape, timing is everything. High-stakes product launches, marketing campaigns, and major announcements frequently underperform-not due to poor execution, but because they are launched in noisy, overcrowded time slots without prior visibility. The Event Collision Detector brings predictive intelligence to your release calendar. By continuously aggregating cross-industry benchmarks, competitor movements, and cultural events, we empower teams to claim clear air space and maximize market impact.

2. Problem Statement

- **Blind Spot Scheduling:** Teams plan launches using internal calendars alone, completely blind to external market activity, overlapping competitor reveals, and major cultural distractions.
- **Diluted Audience Attention:** When events collide, media coverage splits, attendee participation plummets, and impression costs surge across every digital platform.
- **Sunk Marketing Spend:** Launching on an unvetted date degrades ROI, squanders months of preparation, and forces teams to compete against noise rather than stand out.

3. Proposed System: Our Innovative Solutions

- **Cross-Industry Event Radar:** Continuously monitors global event databases, industry conferences, competitor launch windows, and major public events to build a live 360-degree timeline.
- **Real-Time Collision Scoring:** Instantly evaluates proposed launch dates against conflicting events, calculating a risk score based on target audience overlap and media friction.

- **Optimal Launch Recommendations:** Leverages predictive analytics to pinpoint quiet, high-visibility scheduling windows, ensuring maximum reach, attendance, and revenue.

4. Objectives

- **Develop a centralized data pipeline:** Build automated web scrapers and API integrations to continuously aggregate global event databases, industry conferences, and cultural events into a Cross-Industry Event Radar .
- **Build the scoring engine:** Implement an algorithm to calculate a Real-Time Collision Score for proposed dates based on target audience overlap and media friction .
- **Create a recommendation system:** Use predictive analytics to pinpoint quiet, high-visibility scheduling windows .
- **Design a user-centric dashboard:** Build a web application where marketing and product teams can input their proposed launch dates and instantly see a risk breakdown.
- **Implement calendar integrations:** Allow users to sync the radar with their internal planning tools (e.g., Google Calendar, Outlook).

5. Scope of the Project

The initial release will focus on a B2B SaaS web application designed for marketing and product teams. It will include user authentication, an interactive calendar dashboard, the event ingestion pipeline, and the core collision scoring system.

5.1 Scope Refinement

To ensure the project is deliverable within a single semester, the system will initially track a limited number of industries (e.g., Tech and Entertainment). We will rely on public APIs (like Eventbrite or PredictHQ) rather than building complex, multi-site web scrapers from scratch. The predictive analytics will focus on heuristic-based scoring rather than deploying a heavy, deep-learning model.

6. System Workflow — Life of a Date Check

1. **Input:** A marketing manager logs in and inputs a proposed product launch date, selecting their specific industry and target audience.
2. **Data Retrieval:** The backend queries the centralized database for any major public events, competitor launch windows, or conferences happening within a 7-day radius of the proposed date.

3. **Risk Evaluation:** The real-time scoring service calculates the risk score by weighing the size of overlapping events against the user's specific audience overlap and media friction .
4. **Display:** The dashboard visually displays the risk score (e.g., High, Medium, Low). If the score is high, the system highlights the conflicting events causing the blind-spot scheduling .
5. **Recommendation:** The recommendation engine scans the surrounding weeks to suggest 2-3 optimal launch dates that offer clear air space .

7. Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS (for an interactive calendar UI)
Backend	Python with FastAPI (excellent for handling data-heavy ML endpoints)
Database	PostgreSQL (user data) and Elasticsearch (for fast querying of dates and events)
Data Ingestion	Apache Airflow (to run nightly jobs pulling from event APIs)
Scoring / ML	Python (Pandas, Scikit-learn) for calculating target audience overlap and media friction
Version Control & CI/CD	Git/GitHub, GitHub Actions

8. Expected Outcome

By the end of the semester, the deliverable will be a functional, deployed web application. Users will be able to log in, propose a launch date, and receive an instant, data-backed collision report. This tool will successfully demonstrate how integrating external market activity can solve the problem of diluted audience attention and sunk marketing spend .